International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

Student Practical Laboratory Manual

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Introduction

This laboratory exercise focused on fundamental computer system concepts relevant to digital forensics. The practical work involved examining system specifications, operating-system information, storage devices, partitions, processes and file systems. Both Windows and Kali Linux environments were used to observe how operating systems represent hardware and storage information. Partition tables and disk sectors were examined to understand how storage is structured and how partition sizes can be calculated manually. The exercise also introduced MBR and GPT partitioning schemes and the relationship between firmware, boot loaders and operating systems. These activities demonstrated how system and storage information can be collected and interpreted as digital forensic evidence.

NB: The legacy WMIC commands were unavailable on the Windows system, so PowerShell/CIM commands were used as read-only alternatives.

Tools Used

Windows 11: system and physical disk examination.

System Information (msinfo32): BIOS mode and system information.

PowerShell: disk, partition, volume and process examination.

Kali Linux: Linux disk and partition examination.

Fdisk: partition-table examination.

Parted: disk and partition information.

Lsblk: block-device and file-system information.

Blkid: file-system and partition identification.

df: mounted filesystem and storage-usage information.

dd and xxd: read-only sector-0 hexadecimal examination.

Microsoft Word: preparation of the laboratory report.

Lab Scenario

You are a junior digital forensic trainee asked to prepare an initial system profile for a computer that may later be used in a forensic investigation. Your task is to document the system safely, identify possible evidence layers, inspect disk and partition details, calculate partition sizes, explain the boot process, and identify file systems without altering the computer.

Part A - System Identification and Evidence Layer Mapping

In this section, you will identify the major computer system layers and explain how each layer can produce forensic evidence.

A1. Capture basic system specification

- On Windows, open System Information by pressing Windows + R, typing msinfo32, and pressing Enter.
- Capture a screenshot showing OS Name, Version, System Type, Processor, BIOS Mode, Installed Physical Memory, and Secure Boot State if available.

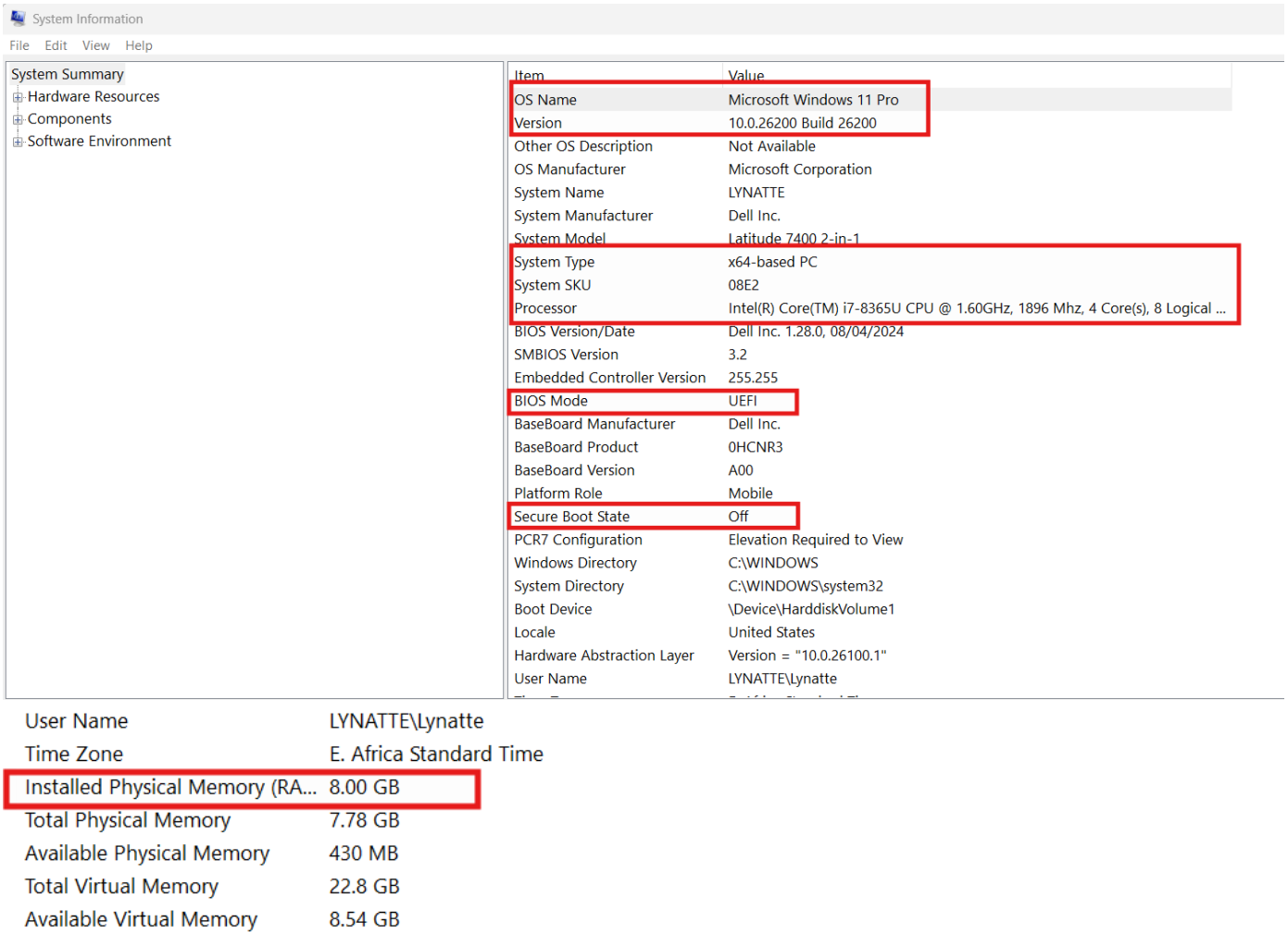


Figure 1: Showing basic system specifications

- Also run the command below in Command Prompt and capture the first page of output.

```
systeminfo
```

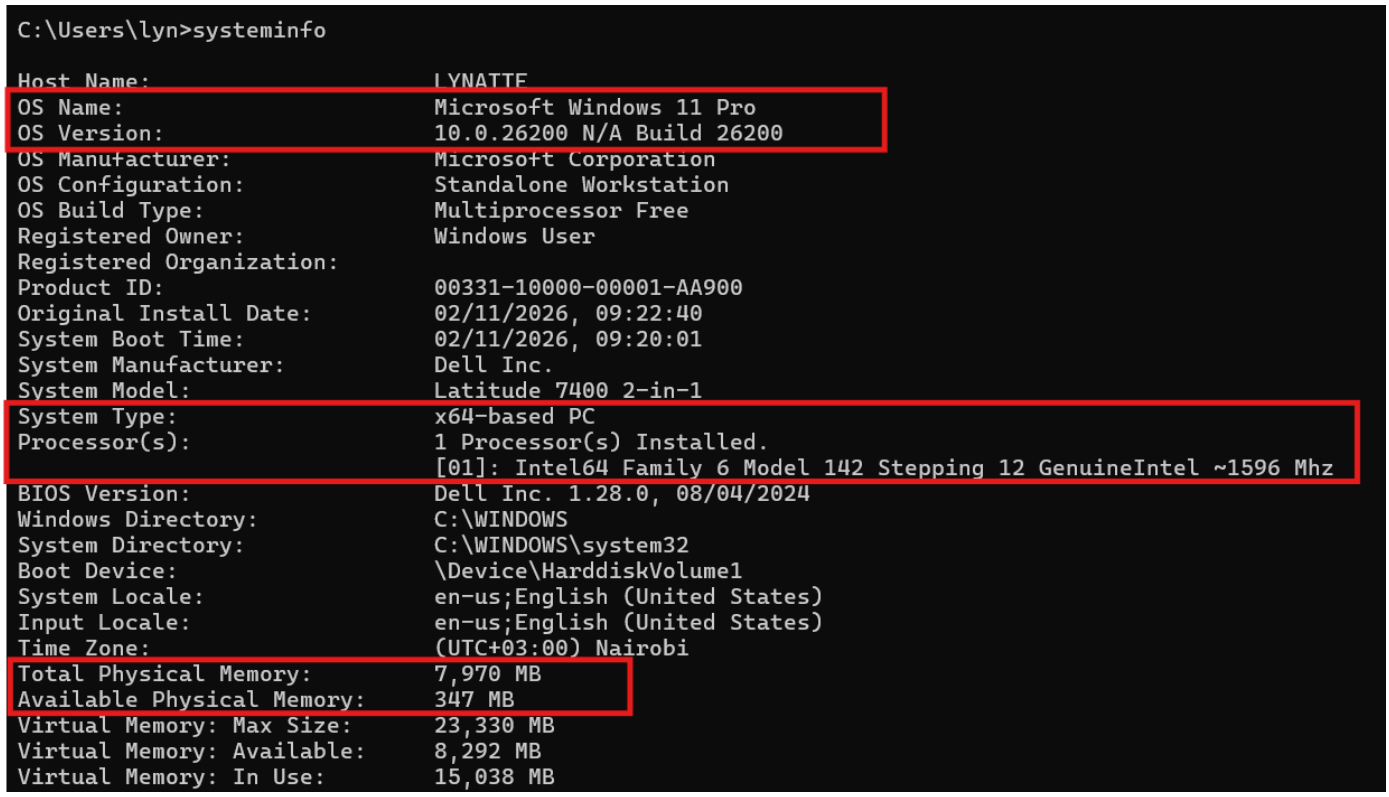


Figure 2: Showing system information in cmd

A2. Capture storage information

- Open Command Prompt or PowerShell.
- Run one of the command sets below. Capture screenshots of storage model, size, partitions, and file systems.
- If a command is not supported on your system, use the alternative command.

```
wmic diskdrive get model,size,serialnumber
wmic logicaldisk get caption,filesystem,size,freespace,volumename

PowerShell alternative:
Get-Disk
Get-Partition
Get-Volume
```



Figure 3: Showing disk storage information in powershell

```
PS C:\WINDOWS\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_sk&prod_hynix#4&26729448&0&000200#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                275.64 GB Basic
4                            296087453696                              1.12 GB Recovery
5                D           297289121792                             199.46 GB Basic
6                            511453429760                               625 MB Recovery
```

Figure 4: Showing partition information in powershell

```
PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C                        NTFS           Fixed     Healthy      OK                    147.3 GB 275.64 GB
                         NTFS           Fixed     Healthy      OK                    610.89 MB    625 MB
D           New Volume   NTFS           Fixed     Healthy      OK                    104.47 GB 199.46 GB
                         NTFS           Fixed     Healthy      OK                    114.57 MB   1.12 GB
```

Figure 5: Showing volume information in powershell

A3. Map evidence sources to system layers

- Create a table in your report showing at least five system layers/sources and the type of evidence that can be generated from each one.
- Use this guide: User activities = Applications + Operating System + Disk/Memory.
- Explain why evidence may exist in more than one layer.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, shell history	Links user action to system activity
Application layer	Office document metadata, browser cache, chat logs	Application folders and databases	Shows what software created or used evidence
OS layer	Event logs, installed programs, user profiles	Windows logs, registry, Linux /var/log	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions	File system and unallocated space	Supports recovery and timeline analysis
Memory/process layer	Running malware, PIDs, network connections	RAM capture, process list	Shows live system behavior

Evidence may exist in more than one system layer because the same user action can generate multiple related artifacts. For example, when a user opens a document, the application may record the document activity, the operating system may generate related access or event records, and the disk may retain the

document and associated metadata. Similarly, browser activity may be represented in browser history at the application layer, system activity at the operating-system layer, and cached or stored data on the disk. Therefore, forensic investigators should correlate evidence from multiple layers rather than relying on a single artifact. This helps establish a more complete and reliable understanding of what occurred on a system.

Part B - Program, Process and Thread Investigation

This section links the slide concept of program, process, and thread to forensic observation.

B1. Observe running processes

- Open Task Manager on Windows, or use Command Prompt/PowerShell.
- Capture a screenshot showing running processes.
- Select three processes and record their process names, PID values, and likely purpose.

```
tasklist

PowerShell alternative:
Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet
```

Process Name	PID	Likely Purpose
chrome	352	Runs a Google Chrome browser process, supporting web browsing and browser-related activities.
backgroundTaskHost	19068	Hosts background tasks for Windows applications.
AggregatorHost	9476	Supports certain Windows background and system aggregation tasks.

```
PS C:\WINDOWS\system32> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                     Id CPU           WorkingSet
----                     -- ---           ----------
AggregatorHost         9476                5332992
ai                     2056 2.078125       1982464
AppActions             5892 21.015625      7647232
backgroundTaskHost    19068 0.625         24514560
bash                   2728 2.421875       9461760
bcmHostStorageService  8048                3092480
bcmUshUpgradeService   6960                2019328
chrome                  352 23729.703125 512159744
chrome                 1056 0.140625       50806784
chrome                 1604 41.140625      19070976
chrome                 2460 5.53125       179081216
chrome                 2852 112280.375  -1769787392
chrome                 2952 0.109375       36880384
chrome                 4540 0.125          51425280
chrome                 7084 0.109375       51347456
chrome                 8060 3.53125         2662400
chrome                 8824 3520.4375     184766464
chrome                 9304 12.78125        6184960
chrome                11500 0.90625        21475328
chrome                12968 51.046875      22274048
```

Figure 6: Showing running processes

B2. Explain the difference

- In your report, explain the difference between a program, a process, and a thread in your own words.

A program is a collection of instructions written to perform a particular task. It is stored on a storage device and does not necessarily have to be running. For example, Google Chrome installed on a computer is a program.

A process is an instance of a program that is currently executing. When a user opens Google Chrome, the operating system creates one or more processes to run it. Each process is assigned a unique Process Identifier (PID) and is allocated resources such as memory and CPU time.

A thread is the smallest unit of execution within a process. A process can contain multiple threads, allowing different tasks to be performed concurrently. For example, a browser may use different threads to handle user interactions, load web pages and perform background operations.

Explain why process information may be useful during malware analysis or live forensic triage.
Process information is useful during malware analysis and live forensic triage because it helps investigators understand what is currently running on a computer. Examining process names, PIDs, CPU usage, memory consumption and executable paths can help identify unusual or potentially malicious activity.
For example, an unfamiliar process consuming unusually high system resources or executing from a suspicious directory may require further investigation. Investigators can also examine parent-child process relationships, loaded modules and network connections to identify suspicious execution patterns.
Process information is particularly valuable during live forensic investigations because it provides insight into the computer's current state. This can help investigators identify running malware, suspicious applications and potentially compromised processes before the computer is shut down and volatile evidence is lost.

Part C - Disk Sector and Disk Size Calculation

Key formula
Disk or partition size = number of sectors x bytes per sector. To convert bytes to GB, divide by 1024 x 1024 x 1024.

C1. Reproduce the slide calculation

Use the values from the slide to calculate the total disk size: sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes

Item	Formula / Value	Answer
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

C2. Calculate your own disk or partition size

- Use your partition table or disk details to identify at least two partitions.
- For each partition, record the start sector, number of sectors, sector size, and file system where available.
- Calculate the size of each partition in GB using the formula above.
- Your calculation must be shown step-by-step. Screenshots alone are not enough.

Example calculation:
Partition size in GB = sector_count x 512 / 1024 / 1024 / 1024
If sector_count = 83,881,984:
83,881,984 x 512 / 1024 / 1024 / 1024 = 39.99 GB

```
PS C:\WINDOWS\system32> Get-Disk

Number Friendly Name                                  Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                                                       Style
------ -------------                                  -------------                  ------------         -----------------      ---------- ----------
0      SK hynix SC401 SATA 512GB                      MI94T002811604H2S              Healthy              Online                  476.94 GB GPT


PS C:\WINDOWS\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_sk&prod_hynix#4&26729448&0&000200#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                                  Size Type
---------------  ----------- ------                                                  ---- ----
1                            1048576                                              100 MB System
2                            105906176                                             16 MB Reserved
3                C           122683392                                         275.64 GB Basic
4                            296087453696                                        1.12 GB Recovery
5                D           297289121792                                      199.46 GB Basic
6                            511453429760                                        625 MB Recovery


PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C                        NTFS           Fixed     Healthy      OK                    147.56 GB 275.64 GB
                         NTFS           Fixed     Healthy      OK                    610.89 MB    625 MB
D           New Volume   NTFS           Fixed     Healthy      OK                    104.47 GB 199.46 GB
                         NTFS           Fixed     Healthy      OK                    114.57 MB   1.12 GB
```

Figure 7: Showing disk,partition and volume information

```
PS C:\WINDOWS\system32> Get-Partition | Format-Table DiskNumber,PartitionNumber,DriveLetter,Offset,Size,Type

DiskNumber PartitionNumber DriveLetter       Offset         Size Type
---------- --------------- -----------       ------         ---- ----
         0               1                   1048576    104857600 System
         0               2                 105906176     16777216 Reserved
         0               3           C     122683392 295964770304 Basic
         0               4            296087453696   1200619520 Recovery
         0               5           D 297289121792 214163259392 Basic
         0               6            511453429760    655360000 Recovery


PS C:\WINDOWS\system32> Get-Volume | Format-Table DriveLetter,FileSystem,FileSystemLabel,Size,SizeRemaining

DriveLetter FileSystem FileSystemLabel         Size SizeRemaining
----------- ---------- ---------------         ---- -------------
          C NTFS                       295964766208  158445318144
            NTFS                          655355904     640561152
          D NTFS       New Volume      214163255296  112175939584
            NTFS                         1200615424     120131584


PS C:\WINDOWS\system32> Get-Disk | Select-Object Number,FriendlyName,LogicalSectorSize,PhysicalSectorSize,PartitionStyle


Number             : 0
FriendlyName       : SK hynix SC401 SATA 512GB
LogicalSectorSize  : 512
PhysicalSectorSize : 4096
PartitionStyle     : GPT
```

Figure 8: Showing actual partition values

Calculating the two main partitions

Partition 3 (C:)

Item	Value
Disk number	0
Partition number	3
Drive letter	C:
Start offset	122,683,392 bytes
Partition size	295,964,770,304 bytes
Filesystem	NTFS
Logical sector size	512 bytes

Step 1: Calculate the starting sector

122,683,392 / 512 = 239,616

Therefore: Starting sector = 239,616

Step 2: Calculate the number of sectors

295,964,770,304 / 512 = 578,056,192

Number of sectors = 578,056,192

Step 3: Calculate partition size in GB (using the lab's 1024-based conversion)

578,056,192 x 512 / 1024^3

= 275.61GiB

Partition 5 (D:)

Item	Value
Disk number	0
Partition number	5
Drive letter	D:
Start offset	297,289,121,792 bytes
Partition size	214,163,259,392 bytes
Filesystem	NTFS
Logical sector size	512 bytes

Step 1: Calculate the starting sector

297,289,121,792 / 512 = 580,642,816

Starting sector = 580,642,816

Step 2: Calculate the number of sectors

214,163,259,392 / 512 =418,287,616

Number of sectors = 418,287,616

Step 3: Calculate partition size in GB

418,287,616 x 512 / 1024^3

= 199.31GiB

Summary of the results

Item	Partition 3 (C:)	Partition 5 (D:)
Start sector	239,616	580,642,816
Number of sectors	578,056,192	418,287,616
Sector size	512 bytes	512 bytes

Filesystem	NTFS	NTFS
Calculated size	275.61 GiB	199.31 GiB

Partition table evidence

```
PS C:\WINDOWS\system32> Get-Partition | Format-List DiskNumber,PartitionNumber,DriveLetter,Offset,Size,Type


DiskNumber      : 0
PartitionNumber : 1
DriveLetter     :
Offset          : 1048576
Size            : 104857600
Type            : System

DiskNumber      : 0
PartitionNumber : 2
DriveLetter     :
Offset          : 105906176
Size            : 16777216
Type            : Reserved

DiskNumber      : 0
PartitionNumber : 3
DriveLetter     : C
Offset          : 122683392
Size            : 295964770304
Type            : Basic

DiskNumber      : 0
PartitionNumber : 4
DriveLetter     :
Offset          : 296087453696
Size            : 1200619520
Type            : Recovery

DiskNumber      : 0
PartitionNumber : 5
DriveLetter     : D
Offset          : 297289121792
Size            : 214163259392
Type            : Basic
```

```
DiskNumber      : 0
PartitionNumber : 6
DriveLetter     :
Offset          : 511453429760
Size            : 655360000
Type            : Recovery
```

Figure 9: Showing partition table evidence

Part D - Safe Partition Table Inspection

Choose either Windows or Linux. You may do both for extra practice. You must not modify the disk.

D1. Windows method - Disk Management, PowerShell and DiskPart

- Open Disk Management by pressing Windows + R, typing diskmgmt.msc, and pressing Enter.

Capture a screenshot showing your disk layout and visible partitions.

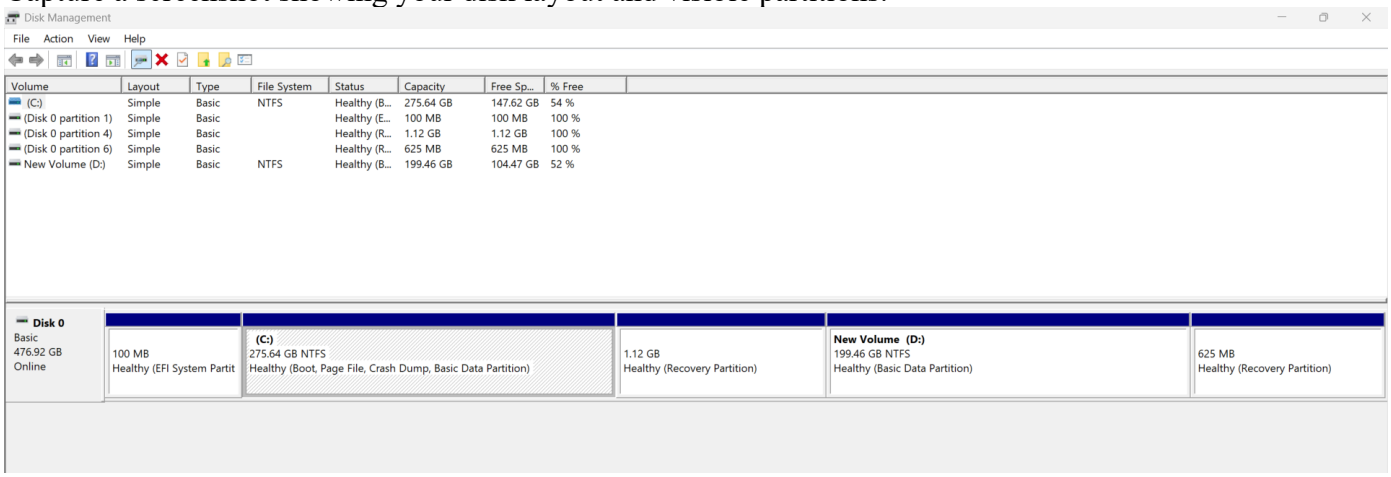


Figure 10: Showing disk layout and partitions

Open PowerShell as normal user or administrator if required and run the safe read-only commands below.

```
PS C:\WINDOWS\system32> Get-Disk

Number Friendly Name                   Serial Number                 HealthStatus         OperationalStatus      Total Size Partition
                                                                                                                   Style
------ -------------                   -------------                 ------------         -----------------      ---------- ----------
0      SK hynix SC401 SATA 512GB       MI94T002811604H2S             Healthy              Online                  476.94 GB GPT
```

Figure 11: Showing inspection of disks

```
PS C:\WINDOWS\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_sk&prod_hynix#4&26729448&0&000200#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                275.64 GB Basic
4                            296087453696                              1.12 GB Recovery
5                D           297289121792                             199.46 GB Basic
6                            511453429760                               625 MB Recovery
```

Figure 12: Showing inspection of partitions

```
PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C                        NTFS           Fixed     Healthy      OK                    147.61 GB 275.64 GB
                         NTFS           Fixed     Healthy      OK                    610.89 MB    625 MB
D           New Volume   NTFS           Fixed     Healthy      OK                    104.47 GB 199.46 GB
                         NTFS           Fixed     Healthy      OK                    114.57 MB   1.12 GB
```

Figure 13: Showing inspection of volumes

- Open Command Prompt as administrator and run DiskPart only for listing disks and partitions. Do not run create, delete, clean, format, active, or convert commands.

PowerShell:
Get-Disk
Get-Partition
Get-Volume

DiskPart safe listing only:
diskpart
list disk
select disk 0
list partition
exit

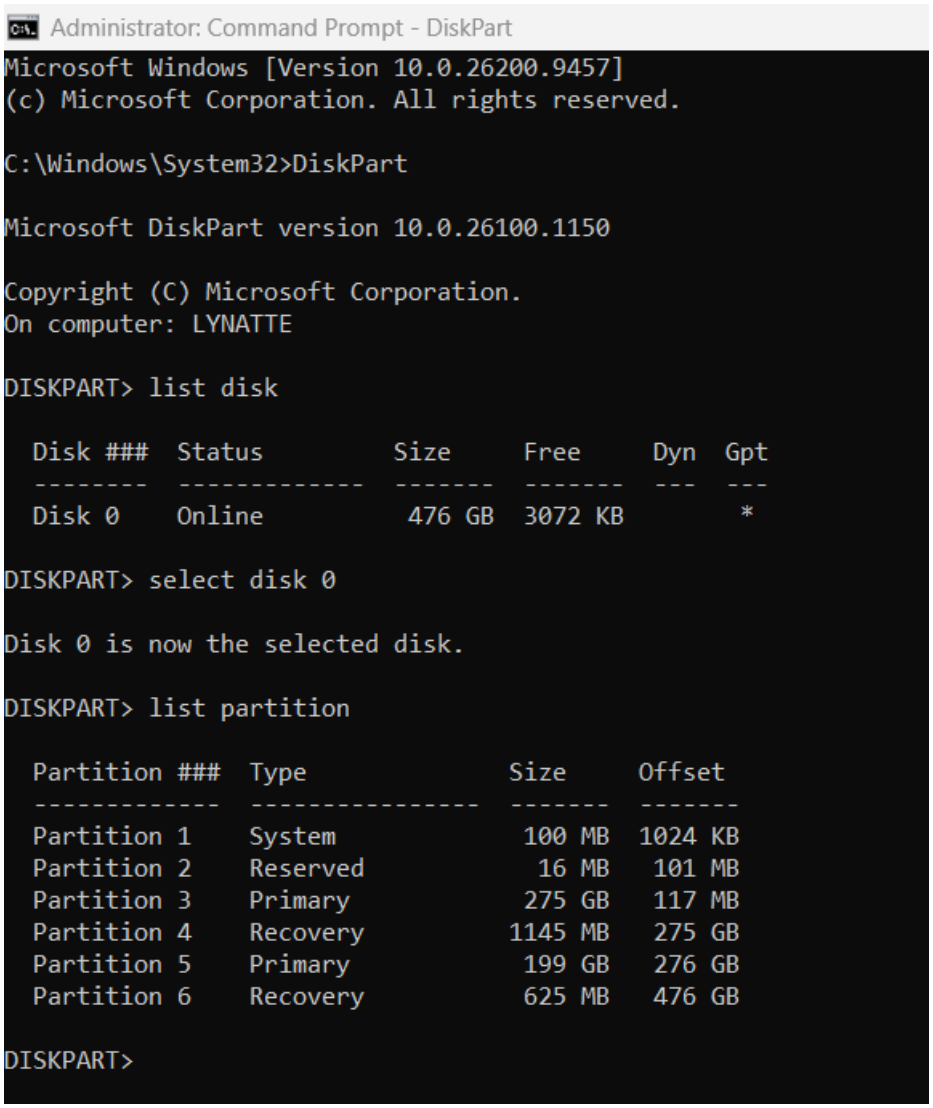


Figure 14: Showing disk and partitions inspections

Disk and partition inspection was performed on the Windows computer using Disk Management, PowerShell and the DiskPart utility. These tools were used in read-only mode to examine the physical storage device, its partition structure and the available volumes without making changes to the disk.

The computer contains one physical disk, identified as an SK hynix SC401 SATA 512GB SSD. The disk has a GPT partition style and contains six partitions, including a system partition, Microsoft Reserved partition, primary data partitions and recovery partitions.

PowerShell commands such as Get-Disk, Get-Partition and Get-Volume were used to obtain information about disk capacity, partition offsets, partition sizes, drive letters and filesystem types. The inspection identified NTFS as the filesystem used by the C: and D: volumes.

This information is relevant to digital forensics because understanding the disk layout helps investigators identify where data is stored, locate relevant partitions and determine which filesystems and structures should be examined during an investigation.

D2. Linux/Kali method - read-only inspection

- Open Terminal in your Linux/Kali VM or host.
- Run the commands below and capture screenshots showing disk, partition, size, and file system information.
- Do not run mkfs, fdisk write commands, parted write commands, dd write commands, or mount read-write commands for this lab.

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
sudo fdisk -l
sudo parted -l
blkid
df -h
```

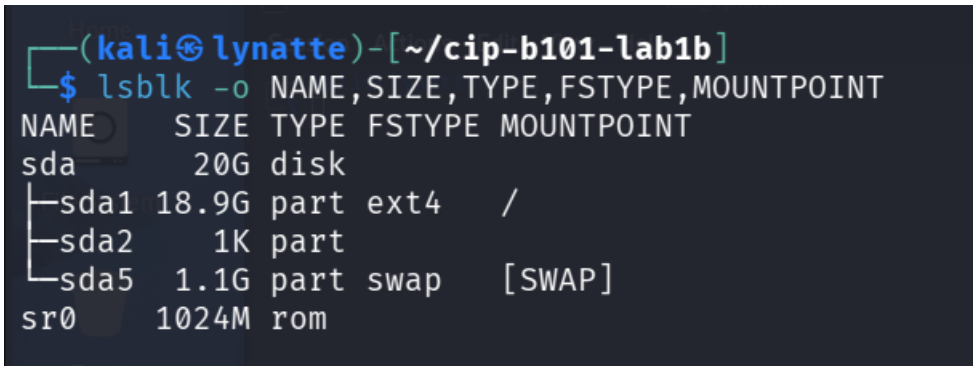


Figure 15: Showing block devices in kali using lsblk

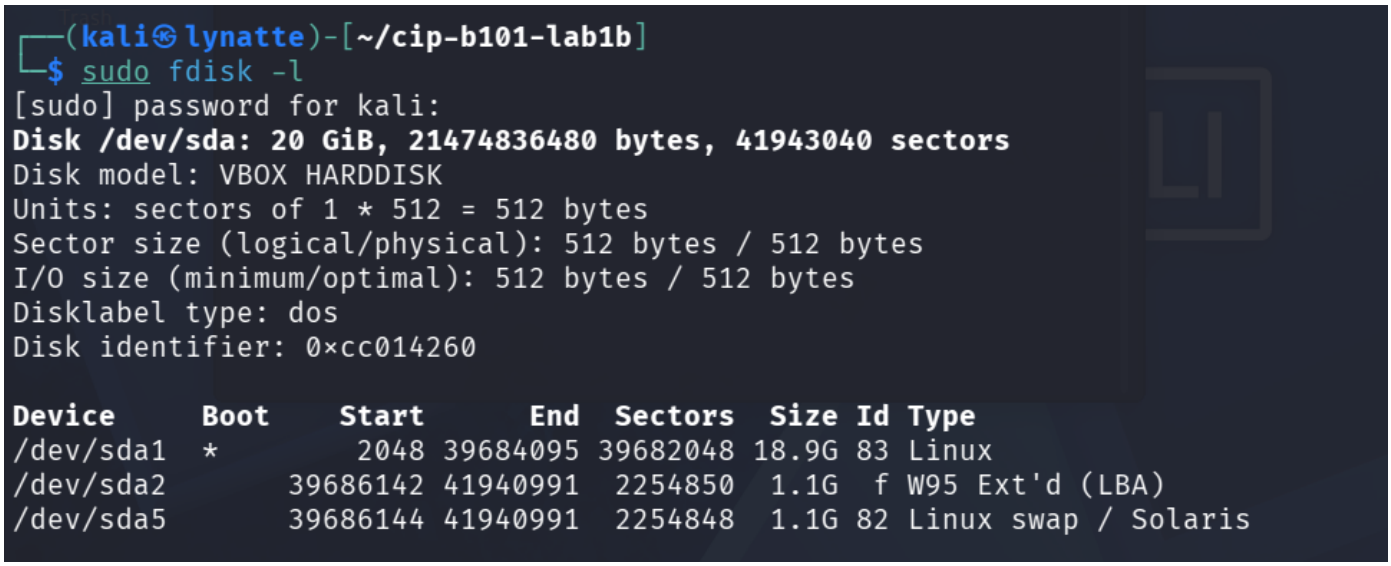


Figure 16: Showing the partition table



Figure 17: Showing inspection of partition information



Figure 18: Showing filesystem types

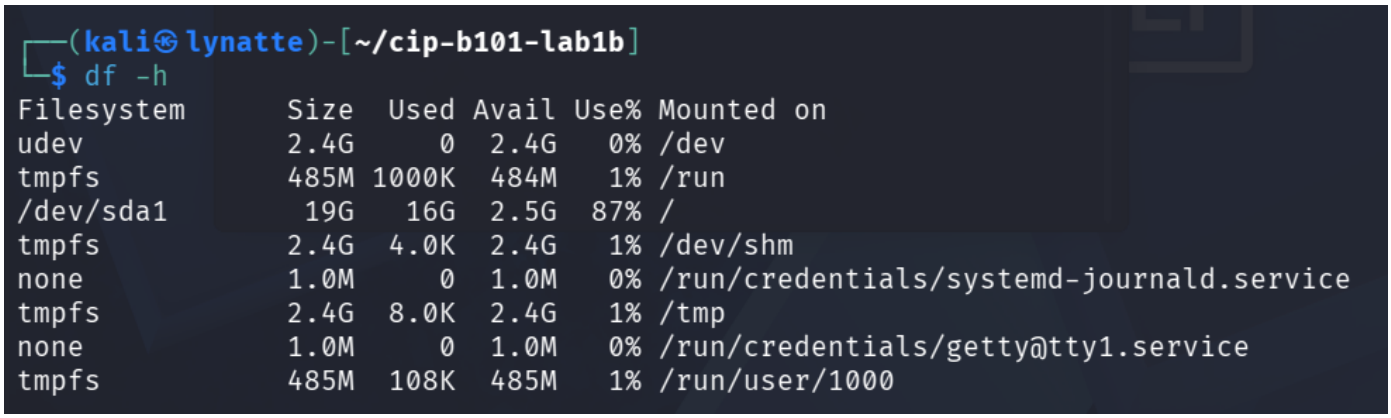


Figure 19: Showing inspection of mounted filesystems

Disk and partition inspection was also performed using the Kali Linux virtual machine. The commands lsblk, fdisk -l, parted -l, blkid and df -h were used to examine the virtual storage device and its partition structure.

The inspection identified one virtual hard disk, /dev/sda, with a capacity of 20 GiB and a sector size of 512 bytes. The disk uses an MBR (msdos) partition table and contains an ext4 primary partition mounted as the root filesystem (/), an extended partition and a logical partition configured as Linux swap.

The blkid command identified the filesystem types and UUIDs, while df -h displayed the capacity and available storage of mounted filesystems. These commands provided complementary information about the disk structure, partition allocation and filesystem configuration.

All inspection commands were used for read-only analysis. No partitions were created, deleted, formatted or modified during the exercise.

D3. Optional WinHex or forensic viewer method

- If your instructor permits, open the physical disk or disk image in a forensic/hex viewer in read-only mode.
- Navigate to sector 0 and capture the first sector where the MBR/protective MBR may be visible.

Identify whether the disk appears to use MBR or GPT. If you cannot safely view this, skip the optional step.

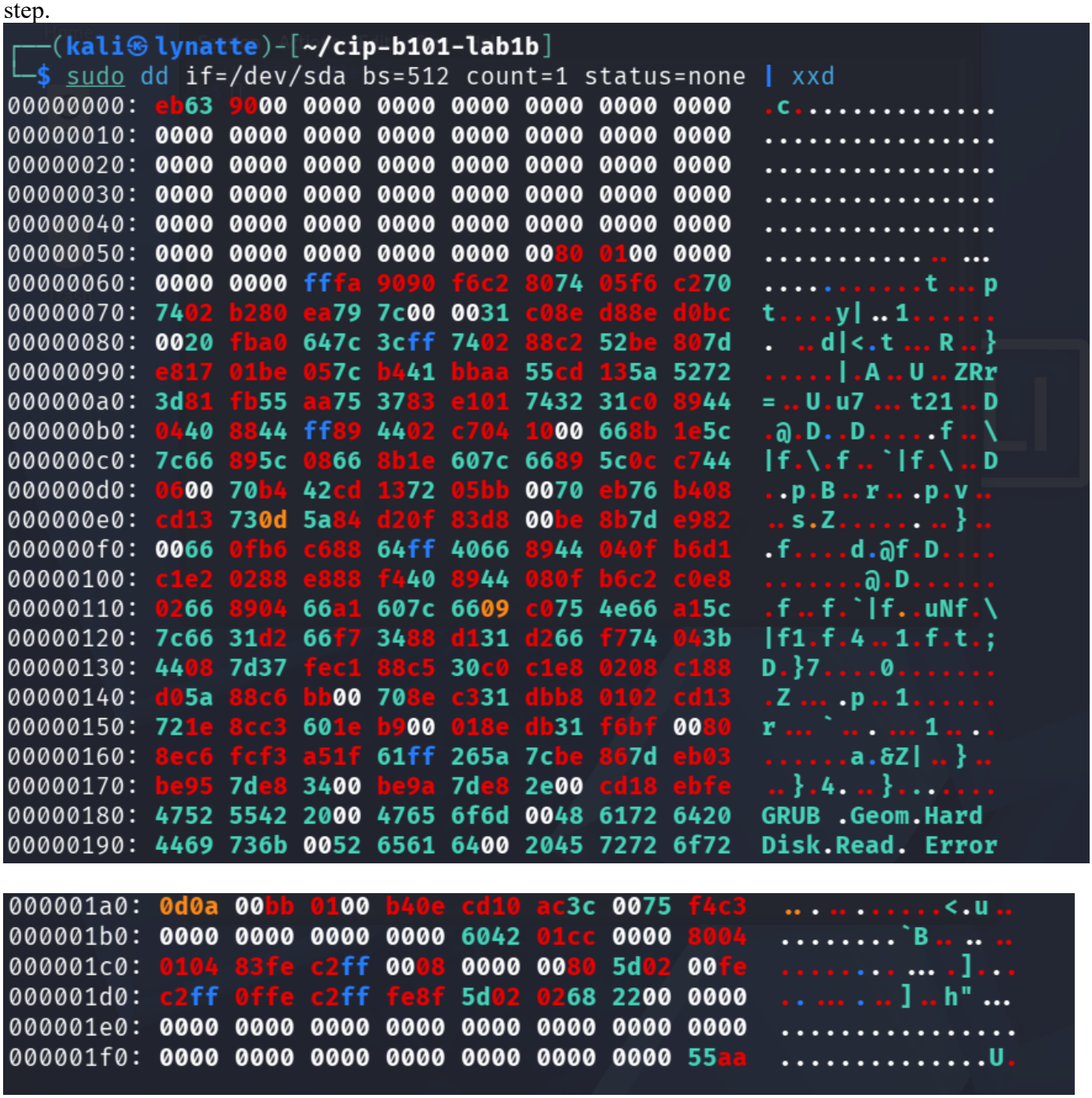


Figure 20: Showing Sector0_Hex_Inspection

A read-only examination of sector 0 of the Kali Linux virtual disk (/dev/sda) was performed using the dd and xxd commands. The command sudo dd if=/dev/sda bs=512 count=1 status=none | xxd was used to read and display the first 512 bytes of the disk in hexadecimal format.

The examination identified the MBR structure. The final two bytes of sector 0 contained the signature 55 AA, located at byte offsets 510–511. The boot code area also contained identifiable GRUB-related strings, indicating the presence of GRUB bootloader-related code.

The partition table begins at byte offset 446 and contains four possible partition entries. The first entry identifies a Linux partition (type 83), while the second identifies an extended LBA partition (type 0F). These findings correspond with the partition information previously obtained using fdisk -l, which identified /dev/sda1 as the primary Linux partition and /dev/sda2 as the extended partition containing the logical swap partition /dev/sda5.

Based on the sector-level examination and the partition table information, the Kali virtual disk uses the MBR (DOS) partitioning scheme. The examination was conducted using read-only commands, and no changes were made to the disk.

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

- Create a simple flow diagram or numbered list explaining how a PC boots.
- Your explanation must include power-on, POST, BIOS/UEFI, CMOS/firmware settings, MBR/GPT or EFI boot records, boot loader/manager, kernel loading, and operating system control.
- State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.

Stage	What Happens	Forensic Relevance
Power-on and POST	Hardware is initialized and checked.	Hardware failure or tampering may affect evidence access.
BIOS/UEFI firmware	Boot device order and firmware settings are applied.	Boot mode can explain partition style and boot artifacts.
MBR/GPT or EFI boot	System locates bootable partition or EFI boot entry.	Boot records may contain evidence of boot configuration.
Boot loader/manager	OS selection and pre-kernel loading occurs.	May show multi-boot or hidden OS behavior.
Kernel and OS control	Operating system fully starts.	Logs and user activity begin to accumulate.

Boot mode of my system: UEFI. This was verified using Windows System Information (msinfo32), where the BIOS Mode was shown as UEFI. The physical Windows disk was also identified as using the GPT partition style through PowerShell.

Stage	What Happens	Forensic Relevance
Power-on and POST	When the computer is powered on, the firmware performs the Power-On Self-Test (POST) and initializes essential hardware such as the processor, memory and storage devices.	Hardware failures or tampering at this stage may affect system startup and the ability to access forensic evidence.
BIOS/UEFI firmware	The UEFI firmware initializes hardware, reads firmware configuration settings and determines the boot device according to the configured boot order.	Firmware settings and boot mode can help explain how the system starts and which boot artifacts should be examined.
GPT/EFI boot	On a UEFI/GPT system, the firmware identifies the EFI System Partition (ESP) and loads the appropriate EFI boot program or boot manager.	EFI boot entries and files can provide evidence about boot configuration, operating systems and possible boot-level tampering.
Boot loader/manager	The boot loader or boot manager prepares the operating system for execution and may provide OS-selection options where multiple operating systems are installed.	Bootloader configuration can reveal multi-boot configurations or unexpected boot entries that may be relevant during an investigation.
Kernel loading	The boot loader loads the operating-system kernel and required initial components into memory. The kernel initializes drivers, memory management and other core system functions.	Kernel and driver information can help investigators understand the software environment and identify potentially suspicious components.
Operating system control	After initialization, control is transferred to the operating system, which starts system services, applications and user processes.	Once the OS is running, logs, processes, network connections and user activity begin generating forensic evidence.

Simple flow diagram

Power On → POST → UEFI Firmware → EFI System Partition / Boot Manager → OS Boot Loader → Kernel Loading → Operating System Control

E2. Compare MBR and GPT

Feature	MBR	GPT
Age / design	Older partitioning system from	Modern partitioning system used

Feature	MBR	GPT
	the 1980s.	with UEFI systems.
Size limitation	Commonly limited to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB.
Partition information	Partition table is stored in the first sector with boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Critical information in sector 0; damage can affect partition interpretation.	Backup structures may help recovery and validation.

Feature	MBR	GPT
Age / design	MBR (Master Boot Record) is an older partitioning scheme dating back to the early PC era.	GPT (GUID Partition Table) is a modern partitioning scheme commonly used with UEFI systems.
Size limitation	With traditional 512-byte sectors, MBR commonly supports disks up to about 2 TB.	GPT supports very large disks and is not subject to the traditional 2 TB MBR limitation.
Partition information	The partition table is stored in sector 0 together with the MBR boot code. It supports up to four primary partition entries, with extended partitions used to create logical partitions.	GPT stores a primary GPT header and partition-entry array near the beginning of the disk and maintains backup GPT structures near the end of the disk.
Forensic issue	Sector 0 contains important boot and partition information. Damage or alteration to the MBR can affect booting and the interpretation of the partition structure.	GPT provides primary and backup partition structures and uses integrity checks, which can assist with validation and recovery if one copy is damaged.
Boot relationship	Traditionally associated with BIOS/Legacy boot, although modern systems can have compatibility arrangements.	Commonly associated with UEFI boot, although GPT itself is a partitioning scheme and does not by itself prove the firmware boot mode.
Partition capacity	Limited partition-entry structure; four primary entries are available directly, with extended/logical partitions used to work around this limitation.	Supports many partitions without requiring the MBR extended/logical partition mechanism.

During the practical investigation, two different disk configurations were observed. The Windows physical computer uses GPT, as confirmed by PowerShell Get-Disk, and its firmware mode is UEFI, as confirmed through msinfo32. The Kali Linux virtual disk uses the MBR/DOS partitioning scheme, as confirmed by fdisk -l, parted -l, and the sector-0 hexadecimal examination. The Kali MBR also contained the 55 AA signature and GRUB-related boot code.

Part F - File System Identification and Interpretation

- Identify the file system used by at least two volumes or partitions on your system.
- For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.
- For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system.
- In your report, explain why a file system is important in digital forensics.

Windows:
wmic logicaldisk get caption,filesystem,size,freespace,volumename
Get-Volume

Linux:
lsblk -f
blkid
df -Th

File System	Common OS / Usage	Forensic Importance
FAT/FAT32	Removable media and older systems	Simple structure; often used on USB drives.
NTFS	Modern Windows systems	Includes metadata such as MFT entries, timestamps, permissions, and journals.
ext family	Linux systems	Uses inode-based metadata and supports forensic timeline analysis.
XFS/JFS/btrfs	Linux/Unix-like systems	Advanced features may affect recovery and metadata interpretation.
HFS+	Older macOS systems	Important in Mac forensic investigations; slide notes date support limitations.

The file system of a storage volume determines how files, directories, metadata and available storage space

are organized and managed. Identifying the file system is important in digital forensics because different file systems store and manage metadata differently, which affects how an investigator interprets timestamps, deleted files, permissions, journals, allocation information and other evidence.

File System Identification

File System	Common OS / Usage	Forensic Importance
NTFS	Modern Windows systems	NTFS contains important forensic metadata, including Master File Table (MFT) records, timestamps, file attributes, permissions and journaling information. This metadata can assist with file-system analysis and timeline reconstruction.
FAT/FAT32	Removable media and older systems	Has a relatively simple file-system structure and is commonly encountered on USB drives and other removable media. Its allocation information and directory structures can be examined during forensic analysis.
ext4	Linux systems	Uses inodes and other file-system metadata structures to store information about files. Metadata and timestamps can be useful for forensic examination and timeline analysis.
XFS/JFS/btrfs	Linux and other Unix-like systems	These file systems provide advanced storage and metadata features. Their specific structures and features must be understood when recovering files and interpreting forensic evidence.
HFS+	Older macOS systems	Important when investigating older Mac systems because its metadata and file-system structures can contain evidence relevant to forensic investigations.
swap	Linux systems; used for virtual memory	Swap is not a normal file system for storing user files. However, it may contain remnants of processes or data that were previously held in RAM, making it potentially relevant when examining volatile-data remnants.

My System Observations

The Windows physical computer was found to contain NTFS volumes. Get-Volume identified both the C: and D: volumes as NTFS. The Kali Linux virtual disk was also examined, and blkid and lsblk identified /dev/sda1 as ext4 and /dev/sda5 as Linux swap.

These results demonstrate that different operating systems can use different file systems and that the file system must be identified before interpreting storage evidence.

Why File-System Identification Is Important in Digital Forensics

File-system identification is important because the structure and metadata of a file system determine how forensic evidence is located and interpreted. For example, NTFS provides MFT records, timestamps, permissions and journal information that can assist in reconstructing file activity. Linux ext4 uses inode-based metadata that can also provide information about files and their activity. Understanding the file system therefore helps an investigator correctly locate files, examine metadata, identify deleted or unallocated data and construct an accurate forensic timeline.

```
PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C                        NTFS           Fixed     Healthy      OK                    147.61 GB 275.64 GB
                         NTFS           Fixed     Healthy      OK                    610.89 MB    625 MB
D           New Volume   NTFS           Fixed     Healthy      OK                    104.47 GB 199.46 GB
                         NTFS           Fixed     Healthy      OK                    114.57 MB   1.12 GB
```

Figure 21: Showing the inspection of volumes

```
┌──(kali㉿lynatte)-[~/cip-b101-lab1b]
└─$ sudo blkid
[sudo] password for kali:
/dev/sda5: UUID="03a46bf4-10db-4ad1-99c7-251c34afbf67" TYPE="swap" PARTUUID="cc014260-05"
/dev/sda1: UUID="ded40c2e-89ff-4449-ba27-ca93c41f6e5f" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="cc014260-
01"
```

Figure 22: Showing information on disks and partitions

Part G - Final Practical Assignment

Main assignment

Verify the size of partitions 1 and 2 of a PC using a partition table. Record the verification process step-by-step. Calculate the size of each partition in GB. No full credit will be awarded if you do not use partition table evidence.

Task	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot of partition table and command output	Using the sudo fdisk -l partition table, partition 1 was identified as /dev/sda1. The partition starts at sector **2048** and ends at sector **39684095**, with a total of **39,682,048 sectors**. The disk uses a sector size of **512 bytes**.
Record partition 1 sector count	Start sector, end sector or sector count	/dev/sda1 Start sector: 2048 End sector: 39684095 Sector count: 39682048
Calculate partition 1 size	Formula and result in GB	The partition size was calculated from the partition table using:

		Partition size = Sector count × Sector size = 39,682,048 × 512 = **20,317,208,576 bytes** Converting to decimal gigabytes: 20,317,208,576 ÷ 1,000,000,000 = 20.32 GB Using binary units, the result is approximately 18.92 GiB, which corresponds closely to the 18.9G reported by fdisk.
Identify partition 2	Screenshot of partition table and command output	Using the same fdisk -l partition table, partition 2 was identified as /dev/sda2. It starts at sector **39686142** and ends at sector **41940991**, with a total of **2,254,850 sectors**. It is identified as an extended LBA partition.
Record partition 2 sector count	Start sector, end sector or sector count	/dev/sda2 Start sector: 39686142 End sector: 41940991 Sector count: 2254850
Calculate partition 2 size	Formula and result in GB	The partition size was calculated using: **Partition size = Sector count × Sector size** = 2,254,850 × 512 = **1,154,483,200 bytes** Converting to decimal gigabytes: 1,154,483,200 ÷ 1,000,000,000 = **1.15 GB** Using binary units, the result is approximately **1.08 GiB**, which corresponds closely to the **1.1G** reported by fdisk

Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	The calculated size of /dev/sda1 was approximately **18.92 GiB**, while lsblk reported the partition as **18.9G**. These values are consistent, with the small difference being due to rounding and the display of storage sizes in binary units. For /dev/sda2, the partition-table calculation gave approximately **1.08 GiB**, while lsblk displayed **1K**. This apparent difference should not be interpreted as the partition actually being 1 KB. /dev/sda2 is an **extended partition/container**, and lsblk does not display its full container size in the same way it displays the size of the filesystem-bearing partitions. The partition table reported by fdisk is therefore used as the authoritative evidence for the actual extent of /dev/sda2.
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	For /dev/sda1, the calculated size of approximately **18.92 GiB** is consistent with the **18.9G** reported by lsblk. The small difference is due to rounding and the difference between decimal GB and binary GiB units. For /dev/sda2, the calculated size of approximately **1.08 GiB** differs from the **1K** displayed by lsblk. This is because /dev/sda2 is an extended partition/container, and lsblk does not display its full container capacity in the same way it displays a normal filesystem partition. The partition table obtained with fdisk provides the actual start sector, end sector and

		sector count for /dev/sda2, so it is the appropriate evidence for calculating its physical partition size. Therefore, the difference is primarily due to how the extended partition is represented by the OS utility, rather than lost space or filesystem overhead.

Partition 1 and 2 screenshot

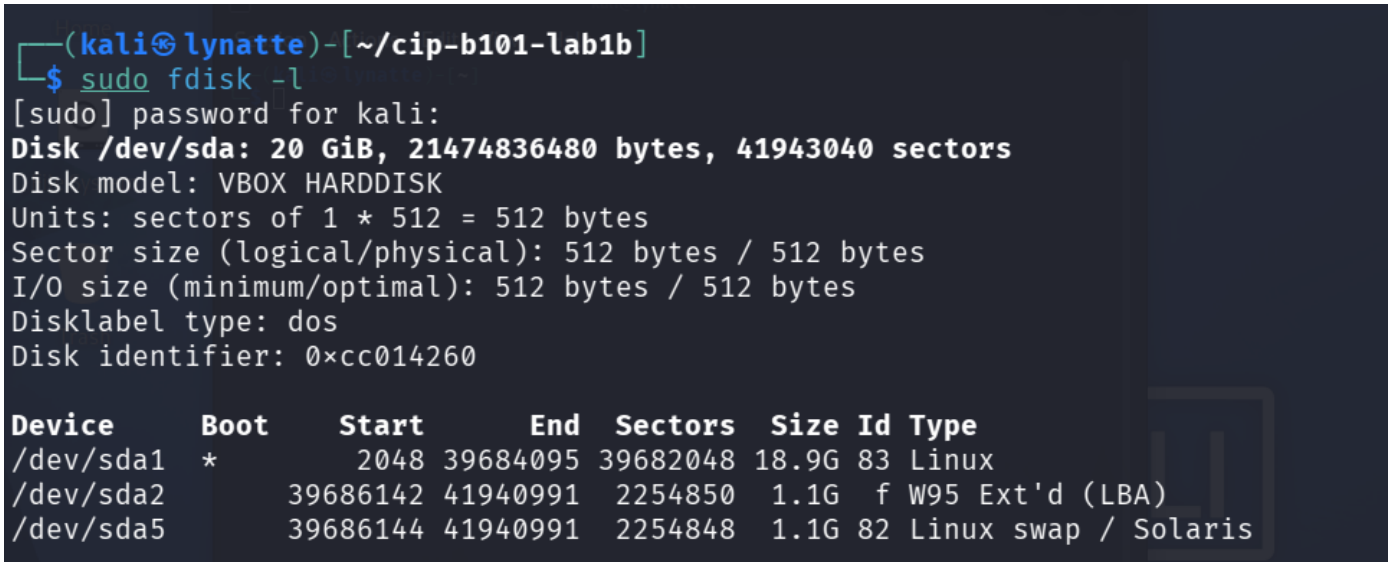


Figure 23: Showing the partition table

Partition 1:
39,682,048 × 512
= 20,317,208,576 bytes
= 20.32 GB
= 18.92 GiB

Partition 2:
2,254,850 × 512
= 1,154,483,200 bytes
= 1.15 GB
= 1.08 GiB

```
┌──(kali㉿lynatte)-[~/cip-b101-lab1b]
└─$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME    SIZE TYPE FSTYPE MOUNTPOINT
sda      20G disk
├─sda1 18.9G part ext4   /
├─sda2    1K part
└─sda5  1.1G part swap   [SWAP]
sr0    1024M rom
```

Figure 24: Showing lsblk

Conclusion

This laboratory provided practical experience in examining computer system architecture, processes, storage devices, partition tables and file systems from a digital-forensics perspective. The investigation demonstrated how Windows PowerShell can be used to obtain system and disk information and how Linux tools such as fdisk, parted, lsblk and blkid can be used to examine storage structures. Manual calculations were performed using sector counts and sector sizes to verify partition capacities. The practical examination also demonstrated the differences between GPT and MBR partitioning, including a sector-level examination of the Kali MBR. In addition, the relationship between UEFI, boot loaders, operating-system kernels and file systems was examined. These skills are important in digital forensics because accurate interpretation of system and storage structures is necessary when identifying, preserving and analysing digital evidence.